

PulseNet-GIS: Project Progress Report

1. Project Overview & Final Vision

PulseNet-GIS is envisioned as a comprehensive emergency medical management platform. The final system architecture dictates an open and secure ecosystem containing:

- Open Registration Platform: Hospitals, Primary Health Centres (PHCs), and Ambulance services register by submitting authenticated information and necessary operational documents.
- Admin Verification Portal: A centralized, secure dashboard at a dedicated URL where administrators can monitor all platform activities and review incoming registration requests (Accept/Reject).
- Secure Access Control: Verified entities receive secure credentials (username/password) to access role-specific dashboards.
- Intelligent Dispatch & Referral Algorithm: A robust, automated system managing patient referrals, resource locking, and nearest-ambulance dispatch.

2. Current Progress (Prototype Status)

To validate the core functionality, a functional prototype has been successfully developed and tested using demo environments. The completed modules include:

2.1 Authentication & Access

- Demo Sign-In System: Implemented a functional login page utilizing demo accounts to facilitate testing and debugging of role-based routing.

2.2 Primary Health Centre (PHC) Module

- Dashboard Interfaces: Developed comprehensive sections including Main Dashboard, Active/Ongoing Referrals, New Referral generation, and Historical logs.

2.3 Hospital Matching & Resource Management

- Targeted Referrals: Integrated logic allowing PHCs to direct referrals to specific, appropriate hospitals based on requirements.
- Resource Locking: Implemented real-time synchronization that locks critical resources (e.g., beds, equipment) at the destination hospital immediately upon referral initiation.

2.4 Ambulance Dispatch System

- Proximity Matching: Successfully implemented logic to identify and alert the nearest available ambulance to the originating PHC.
- Automated Workflow: The system handles the complete dispatch lifecycle - sending notifications, routing the ambulance to the PHC for patient pickup, and navigating to the destination hospital for final handover.

3. Future Roadmap & Next Plans

Based on the current prototype state, the following milestones define the path to finalizing the production-ready PulseNet-GIS platform:

- Develop the Registration Module: Create the frontend forms for Hospitals, PHCs, and Ambulances to

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submit details and upload verification documents.

- Build the Admin Verification Dashboard: Develop a distinct admin portal with capabilities to view all system activities, review submitted registration documents, and securely approve or reject applications.
- Transition to Dynamic Authentication: Deprecate static demo accounts. Implement the automated credential delivery system for approved facilities.
- Enhance Secondary Dashboards: Finalize and polish the specific UI/UX for the Hospital and Ambulance dashboards to match the operational depth of the PHC dashboard.
- Algorithm Refinement & Edge-case Handling: Continuously test the routing and resource-locking algorithms under varied simulated loads to identify and resolve potential issues.